

ASIF RAHMAN

PhD Candidate, Department of Computer Science
The University of Texas at El Paso, El Paso, TX 79968

arahman3@miners.utep.edu · 915-505-0596 · ucchol.github.io
[Google Scholar](#) · [ORCID](#) · [DBLP](#) · [GitHub](#) · [LinkedIn](#)

RESEARCH INTERESTS

Game theory for security, cyber deception, behavioral modeling of attackers, human subjects experiments in cybersecurity, and machine learning for threat detection. My work asks how defenders should plan when the adversary is a bounded human decision maker rather than a perfect optimizer, and increasingly when the adversary is an AI agent.

EDUCATION

The University of Texas at El Paso PhD in Computer Science. Advisor, Christopher D. Kiekintveld.	expected Spring 2028
The University of Texas at El Paso MS in Computer Science	2023
Khulna University of Engineering and Technology, Bangladesh BSc in Electronics and Communication Engineering	2016

PUBLICATIONS

Peer-Reviewed Conference Papers

- [C5] A. Rahman, A. A. N. Ryen, A. Hemida. *Beyond Best Response: Quantal Stackelberg Deception as Insurance Against Attacker Misspecification*. Conference on Decision and Game Theory for Security (GameSec), 2026. **Accepted, oral presentation.**
- [C4] M. A. Sayed, A. Rahman, A. Hemida, C. D. Kiekintveld, C. A. Kamhoua. *Coordinated Multi-Domain Deception: A Stackelberg Game Approach*. International Conference on Computing, Networking and Communications (ICNC), 2026, pp. 85–89. [arXiv:2601.02596](https://arxiv.org/abs/2601.02596).
- [C3] A. Rahman, C. Natividad, M. A. Sayed, P. Aggarwal, C. D. Kiekintveld. *Using Choice Overload to Degrade Cyber Attacks*. Conference on Decision and Game Theory for Security (GameSec), 2025. Lecture Notes in Computer Science, vol. 16224, Springer, pp. 23–42. [doi:10.1007/978-3-032-08067-7_2](https://doi.org/10.1007/978-3-032-08067-7_2). Code, github.com/ucchol.
- [C2] M. A. Sayed, A. Rahman, C. D. Kiekintveld, S. García. *Fine-tuning Large Language Models for DGA and DNS Exfiltration Detection*. Annual Computer Security Applications Conference Workshops (ACSAC Workshops), 2024. [doi:10.1109/ACSACW65225.2024.00034](https://doi.org/10.1109/ACSACW65225.2024.00034). [arXiv:2410.21723](https://arxiv.org/abs/2410.21723).

Preprints

- [P1] T. Ige, C. D. Kiekintveld, A. Piplai, A. Rahman, O. Kolade, S. Kunapuli. *MAD-OOD: A Deep Learning Cluster-Driven Framework for Out-of-Distribution Malware Detection and Classification*. 2025. [arXiv:2512.17594](https://arxiv.org/abs/2512.17594).

Manuscripts in Preparation

- [M1] A. Rahman et al.. *Process models of adversarial choice under information load*. Cognitive process modeling of attacker target selection. In preparation for CHI 2027.

HONORS, AWARDS, AND GRANTS

Partial travel grant, GameSec 2025, Athens, Greece	2025
Research conducted under the IARPA / ODNI ReSCIND program, Contract N66001-24-C-4504, and DEV-COM Army Research Laboratory Cooperative Agreement W911NF-23-2-0012.	

RESEARCH EXPERIENCE

Graduate Research Assistant, The University of Texas at El Paso

2021–present

- Built the research infrastructure for a quantal Stackelberg approach to cyber deception, including a gradient-based solver with analytical Jacobian and multi-start restarts and a 16-node cyber-physical testbed drawn from real CVE records. Proved that the resulting defense behaves as insurance, giving up little value when the attacker model is correct and protecting substantial value when it is misspecified [C5].
- Designed and ran a behavioral experiment on adversarial file selection and consumer choice, then modeled the data with conditional logit, mixed logit, and multi-attribute accumulator models. Attribute detail rather than option count is what breaks the link between stated preferences and observed choice.
- Developed coordinated deception across coupled cyber and physical layers, using CVSS-derived exploit probabilities to place deceptive resources [C4].
- Fine-tuned large language models to detect domain generation algorithms and DNS exfiltration, evaluated across 59 real-world DGA families with emphasis on generalization to unseen families [C2].

TEACHING EXPERIENCE

Graduate Teaching Assistant, The University of Texas at El Paso

2021–present

Operating Systems, Computer Networks, and Machine Learning. Led lab sections, held office hours, and mentored student course projects.

PROFESSIONAL SERVICE

Program Committee, AI for Social Impact (AISI) track, AAAI 2027	2026
Program Committee, AI for Social Impact (AISI) track, AAAI 2026	2025
Reviewer, AAMAS	2025
Reviewer, ICNC	2026
Reviewer, ICECER	2026

TECHNICAL SKILLS

Languages	Python, C++, Java, JavaScript, Bash
Modeling and ML	PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, pandas. LLM fine-tuning with LoRA, discrete choice and accumulator models, clustering
Security	Game-theoretic deception modeling, CVE, CVSS, and NVD vulnerability data, malware and DGA detection, packet analysis
Systems	Linux, Git, Docker, Kubernetes, AWS, PostgreSQL, MySQL

PROFESSIONAL EXPERIENCE

bKash Limited, Bangladesh

2018–2021

Engineer, Service Operations. Ran a hybrid AWS and on-premises production stack for in-app banking, owning incident triage, payment API integration, and Python automation of log parsing and reporting.

REVE Systems Limited, Bangladesh

2016–2018

System Administrator. Maintained production MySQL servers and diagnosed VoIP service disruptions through SIP and UDP traffic analysis.